ArchSync: Evidence-Backed Conformance Checking for TypeScript Service Architectures

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Service changes can introduce architectural drift while preserving functional behavior, leaving maintainers without a clear link between a violated design rule and the code responsible. ArchSync combines an explicit architectural contract, provenance-aware extraction of selected TypeScript service interactions, and a Git gate that distinguishes introduced violations from changes requiring an architecture decision. We investigate implementation correctness using co-developed patches and detector signals, followed by finite rule checking and adversarial testing. All 20 development patches match their declared classifications; targeted adversarial inputs expose extraction and incremental-analysis defects that are repaired and retained as regression cases. The contribution is an executable integration of established conformance concepts with service-level evidence and change-time decisions, supported by inspectable failure and repair records. These development artifacts provide a reproducible basis for evaluating architecture checks at pull-request time.

CCS Concepts: • **Software and its engineering → Software architecture**; *Software verification.*

Additional Key Words and Phrases: architecture conformance, service-oriented systems, static analysis, architecture drift, reproducibility

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1 Introduction

Service-oriented systems can preserve functional behavior while introducing a forbidden dependency, losing a required interaction, or adding infrastructure that needs a design decision. These changes concern different architectural judgments. A useful check must connect an observed interaction to the declared contract, identify supporting source evidence, and distinguish a violation from an evolution. ArchSync addresses this information-extraction and decision-support problem with deterministic rules and inspectable evidence.

Our contribution is an executable integration: (1) a typed expected/observed service graph with explicit constraints; (2) provenance-aware extraction of selected TypeScript interactions and evidence; and (3) a Git-diff gate that reports introduced violations separately from evolution. Building on established model-comparison and dependency-rule

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concepts, the integration connects a service interaction, its supporting source location and its effect on a proposed change. Retained failure and repair records expose how extraction and incremental-analysis choices affect that decision.

We ask whether the prototype reconstructs declared graph changes and detector signals (RQ1), classifies controlled changes correctly (RQ2), supplies the declared evidence locations (RQ3), and preserves results under replay and incremental execution (RQ4). These questions guide the controlled evaluation in Section 4 and the reliability extension in Section 5. For practitioners, the Git-scoped gate provides a concrete pull-request check: introduced findings carry rule identifiers and inspectable evidence, while evolution is routed for an explicit architecture decision. The decision path executes deterministic rules without an LLM inference service.

2 Background and Related Work

This scoped narrative review positions ArchSync against foundational and contemporary conformance approaches using sources selected purposively from the retained keyword-search corpus. Contemporary evidence is from 2022–2025; two older references establish the origins of model comparison and dependency constraints.

Problem and observation boundary. A recent mapping characterizes architecture erosion [8]. Code-review studies identify symptoms in OpenStack and OpenStack/Qt projects [7, 9]; their project overlap limits independence. Practitioner evidence identifies documentation and shared practices as ways to manage drift [1]. A recovery taxonomy and microservice-tool mapping distinguish reconstruction purposes and applicability [2, 12]. View-based drift analysis compares intended and reconstructed specifications in an industrial case [16]. These works motivate an explicit observation boundary between a source graph and the runtime architecture.

Closest conformance approaches. Software Reflexion Models establish high-level/source model comparison [11], and Dependency Constraint Language develops declarative dependency constraints [15]. ArchSync builds on these concepts. CALint checks Python dependencies against Clean Architecture rules [3]. ArchLintor detects TypeScript module-interaction violations, provides source-linked diagnostics and describes CI/CD integration [13]. ArchSync's contribution combines typed service interactions, their source evidence and Git-time introduced-finding decisions. For example, changing a fetch destination can change a service relationship while leaving module imports unchanged. Linking that relationship to a rule and a source location makes the architectural consequence inspectable; comparing base and head distinguishes an introduced violation from a new relation requiring an architecture decision. LLM-generated architectural tests introduce a further rule-translation step [10]; ArchSync executes explicitly supplied deterministic rules.

Change-time feedback and verification. Repository mining studies microservice evolution [4], and industrial conformance visualization reports practitioner experience [17]. ArchSync uses relationship-level keys to keep existing findings visible while directing Git-time decisions to newly introduced findings. Evaluation-method research stresses replicability [6]; interrogation testing shows that targeted inputs can expose analyzer soundness and precision failures [5]. Retained failure/repair evidence makes the implementation of this combined workflow inspectable.

Published evidence and integration value. Table 1 compares the approaches at their reported observation units. ArchLintor illustrates module-level absence and divergence; CALint's publisher abstract reports three cases. Schneider et al. execute nine recovery tools on 17 Java/Spring applications [14], providing a design for recovery evaluation. ArchSync connects selected HTTP, PostgreSQL, Redis and AMQP observations to an explicit service contract and a base/head decision with source evidence. The useful combination is a traceable path from a changed service interaction

Table 1. Published context, not an executed baseline comparison. Different languages, units and datasets prevent numerical ranking.

Approach	Language / focus	Unit and decision	Evaluation evidence
ArchLintor [13]	TypeScript; module conformance	Module dependencies; allowed, forbidden, required	One author-developed eight-module example
CALint [3]	Python; Clean Architecture	Dependencies; dependency rule	Three cases in publisher abstract; full text unavailable
Recovery comparison [14]	Java/Spring; architecture recovery	Components, connections, endpoints; recovery correctness	Nine tools on 17 applications
ArchSync (this work)	TypeScript; service conformance at Git time	Typed service relations; rules and introduced findings	20 co-developed patches; 40 detector signals

to BLOCK or REVIEW, backed by retained regression and failure/repair records. Section 6 defines the scope of these records and the remaining comparative requirement.

3 Executable Contract and Change-Time Analysis

Let $M = (V, E, R)$ be the expected architecture, with components V, directed typed relationships E, and rules R. The observed graph O is reconstructed from supported source constructs. Component identifiers and metadata from M guide reconstruction. Edges collapse repeated observations with the same source, target and kind, while retaining their file/line evidence.

A deny rule rejects matching edges. An allow rule restricts destinations of the selected outgoing relations. A require rule demands a matching direct edge for each selected source; require-path demands a nonempty directed path, optionally restricted to one edge type. An observed rule violation takes classification precedence; otherwise a graph difference is evolution, and an unchanged graph is no-impact. The expected model remains under explicit maintainer control.

Tiny contract example. Let $V = \{w, a, d\}$ denote a web service, API and database, with $E = \{w \xrightarrow{\text{http}} a, a \xrightarrow{\text{data}} d\}$. Let R require a path from w to d with unrestricted edge type. An observation with edge set E satisfies the rule and is no-impact. Removing $a \longrightarrow d$ violates the path requirement; adding $w \xrightarrow{\text{data}} d$ while retaining E is evolution. Against a base matching E, these edits yield BLOCK and REVIEW respectively under the gate below.

The TypeScript adapter recognizes selected global fetch calls and imported PostgreSQL, Redis and AMQP client operations. Import provenance distinguishes package clients from method-name lookalikes. HTTP targets use host identity; AMQP edges capture broker reachability. Evidence scores are heuristic. Section 6 describes the resulting observation boundary.

For a base/head pair, the gate reconstructs the base, analyzes affected source components and merges retained observations with new ones. Full-head scanning supplies a consistency oracle. New violation keys produce BLOCK, new evolution keys produce REVIEW, and no introduced finding produces PASS (Figure 1). Existing findings remain visible. Keys operate at relationship level, so an additional offending call on an existing edge need not introduce a new key. PASS denotes the absence of newly recognized gated findings. REVIEW requires a separate architecture decision; unchanged inputs remain REVIEW after a human rejection.

The expected model supplies the input contract. Call-site evidence points to observed code. A missing required relation has no absent call to localize, so it receives a deterministic component anchor. These evidence types are scored

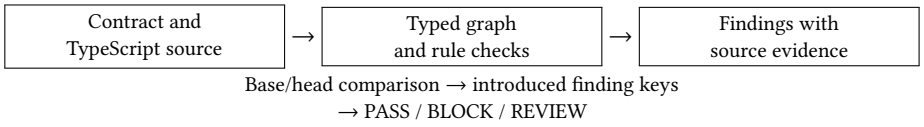


Fig. 1. Git gate flow: introduced violations yield BLOCK; otherwise introduced evolution yields REVIEW, and no introduced finding yields PASS. Maintainers control the contract separately.

Table 2. Implementation agreement with co-developed expectations on one synthetic system.

Measure	Agreement/count
D1 changed nodes / edges	5/5; 12/12
D1 classifications / violation rule sets	20/20; 7/7
Call-site cases / missing-relation anchors	9/9; 2/2
P3 decisions / exact deltas	20/20; 20/20
P3 incremental/full scan / cold–warm replay	20/20; 20/20
P3 parsed / head file instances	57/189

separately. Cache identity binds the baseline and configuration; later hardening also separates repository scope and analyzer version and checks payload integrity. A checksum detects accidental payload corruption.

Implementation artifacts. Core implements model validation, graph comparison and rules; Guardian implements source extraction and the Git gate. Historical measurements use pinned Core (2affbbb0da85) and Guardian (8779bf7965b3) revisions. The packaged source archive and SHA-256 manifest separately identify the later hardening candidate.

Worked controlled example. In D1 case 06, rule `ARCH-001` forbids a direct relationship from `frontend` to `payment-service`. The patch adds `payDirectly` and a fetch call to the payment endpoint. Reconstruction adds one HTTP edge between those components; the deny rule produces an error finding with evidence at `frontend/src/app.ts`, line 14 of the patched file. Because the finding is introduced relative to the base, the Git gate returns BLOCK. A new relation without a violated rule instead produces REVIEW, reserving approval for a separate architecture decision.

4 Controlled Implementation Verification

Objects and independence. D1 is a synthetic Order Platform baseline with five components and five relationships. Its 20 isolated patches comprise nine no-impact changes, seven violations and four evolutions. Topology labels preceded integration, but executable patches and precise evidence locations co-evolved with the analyzer. D2 has four positive and four hard-negative signals for each of five detector groups: fetch, PostgreSQL, Redis, AMQP publish and AMQP consume. It was constructed from v0.1 weaknesses and informed v0.2. P3 replays the same D1 patches through Git.

Procedure and units. D1 validates hashes, patch applicability and expected changes, then runs the baseline and each isolated patch twice (42 executions). D2 runs both pinned versions twice (four executions). P3 creates fresh baseline repositories, applies one patch, performs cold and warm checks, and separately scans the full head. Results are compared after excluding elapsed time and cache-hit state. The 20 cold/warm pairs plus full scans require 80 analyzer calls. D1 items are graph nodes/edges; D2 items are annotated detector–edge–file–line signals, which can share graph edges. Evidence matches use the declared case-level locations.

Results. Table 2 links changed nodes/edges to RQ1, classifications/rule sets to RQ2, call-site/anchor cases to RQ3, and P3 rows to RQ4, retaining the changed-item and case denominators. Full-graph totals also matched 105 nodes and 108 edges, but repeatedly include baseline structure. On D2, v0.1 yielded TP=18, FP=4, FN=2 and TN=16; v0.2 yielded TP=20, FP=0, FN=0 and TN=20. The four false positives were local PostgreSQL/AMQP-shaped receivers; the two misses were Redis operations. These targeted repairs improve the within-tool regression results. Duplicate normalized output matched, and P3 cache miss/hit results agreed in all 20 cases.

The original Windows recording used Node.js 22.16.0 and an Intel Core i7-13650HX. Cold/warm nearest-rank p50 values were 518.51/242.62 ms and p95 values 531.05/249.30 ms. They combine parsing, Git, cache I/O and reporting; RAM, power/load and filesystem-cache state were unrecorded. Historical functional replays passed Linux and Windows, checking installation and repeatability on the same inputs.

Data access. The benchmark repository (24d63ebf2fc3) contains the controlled system, patch labels, packaged runtimes and raw historical records. Section 6 discusses the interpretation of these development results.

5 Separate Technical Reliability Extension

The subsequent 2026-09-26 campaign targets local development defects and preserves original failures. The preceding observations retain their original pinned analyzer packages.

Methods and AI-use disclosure. OpenAI Codex with delegated coding agents assisted in constructing the finite oracle, adversarial inputs and mutation harness, modifying the implementation, executing local checks and inspecting outputs. Reported counts come from retained executable test reports. The development process includes automated cross-agent code review and test refinement after failure inspection, as detailed below.

Finite rule checking. A separately implemented adjacency-matrix and transitive-closure oracle checks all 4,096 directed graphs on three fixed nodes, two edge types and 12 non-self edge slots. Each graph is evaluated against 192 combinations of deny/allow/require/require-path, exact/wildcard selectors and type constraints: 786,432 rule–graph evaluations. Additional checks apply ordering and consistent renaming on 241 graph states and inverse addition/removal deltas on all 4,096.

Test sensitivity. Eight declared implementation mutations target rule enforcement, path traversal and classification. The initial suite killed seven; violation demotion survived because classification was not asserted. Adding the missing assertion produced eight kills, no survivors and no execution errors in the final audit. Both rounds are retained. The two rounds document increased sensitivity to the declared fault model.

Extraction and incremental faults. Adversarial source tests exposed type-only imports treated as runtime clients, protocol mismatches, shadowed bindings and mistaken URL expressions. Binding symbols per file separates local names from package/global names. Later checks exposed wrong-host string concatenation and stale provenance after direct reassignment. Static string composition now resolves the complete URL. A further 14-case extension reproduced 11 failures caused by destructuring, unary updates and loop writes; all 43 cumulative adversarial cases pass after repair. Recognized writes conservatively invalidate the binding throughout its file, including earlier legitimate uses. Unresolved composition emits no relationship.

Incremental checks exposed missed unusual/ignored filenames, nested-component evidence loss, cross-project cache reuse and corrupt-cache behavior. Exact source ownership, Git path handling, repository/version-bound cache identity

and payload checks address the reproduced cases. The extension records failing inputs, repaired outputs and remaining limitations separately. Its full local suites passed 126 Core and 315 Guardian tests on Node.js 22.16.0/macOS arm64. Two inert compiler-host callbacks are excluded from Guardian's configured coverage. The current analyzer is unreleased semantics version 0.4; it also passes a development replay of all 20 D1 cases.

6 Validity and Implications

Construct and internal validity. D1 and D2 are controlled, co-developed datasets; P3 reuses D1. Their perfect matches are regression-oracle agreement. D2 informed detector repairs, so its v0.1/v0.2 comparison is within-tool development evidence, not an external baseline. Repair against known inputs creates an overfitting risk. Shared assumptions between labels, implementation and the separately coded oracle can survive these checks. Handwritten positives and hard negatives emphasize anticipated constructs; their balance does not estimate the prevalence of violations in practice. File/line agreement measures specified case locations rather than complete evidence recall. The typed structural view addresses conformance, leaving responsibilities, runtime quality and practitioner usefulness for separate evaluation.

Analysis and decision boundary. Reconstruction depends on the supplied model and supported source constructs. HTTP host identity collapses port/path distinctions, and AMQP broker reachability leaves routing compatibility unresolved. Dynamic templates, alternative fallbacks, property mutation, imported-constructor rebinding, wrappers and inter-file flows can remain invisible. File-wide binding invalidation can suppress legitimate earlier uses; the extractor is neither sound nor flow-sensitive. Evidence scores are uncalibrated heuristics. Relationship-level keys can hide an additional offending call on an existing edge, and incremental/full agreement can preserve a shared omission. PASS is limited to newly recognized findings; REVIEW leaves approval and rejection enforcement to repository governance. Cache checksums address accidental corruption, not a writer able to replace both payload and checksum.

Reliability scope. The oracle enumerates a finite three-node domain, excluding self-loops, larger graphs, arbitrary wildcard fragments, extraction and component-metadata changes. Its 786,432 combinations are not independent samples. Mutation testing covers eight declared faults and includes a post-failure assertion repair; it is a test-development result rather than a whole-program mutation score. Regression counts describe executed checks. Automated cross-agent review supplies no independent human annotation, adjudication or validation signoff. Historical Linux/Windows replay and the later macOS hardening campaign have distinct execution scopes.

Expected behavior on real projects. In a TypeScript project using supported clients directly and a contract aligned with service boundaries, recognized calls should produce source-linked relations and introduced findings for review. This is an expectation from the extraction design. Wrappers, dependency injection and configuration-built endpoints can hide interactions: a missed forbidden call can falsely PASS, while an edit making a required relation unobservable can introduce a false BLOCK. Shared hosts can merge distinct destinations; file-wide invalidation after a write can suppress valid calls. An outdated contract can flag an intentional migration. Maintainers would need to inspect the evidence and update the model explicitly. Larger repositories may reduce the incremental parsing benefit when changes span many components.

External and comparative validity. No independent repository study was performed. One small synthetic topology cannot represent real dependency diversity, configuration practices, repository scale or change histories. The controlled results therefore leave field accuracy, false-block burden and developer benefit unmeasured. Repeated patches and runs are clustered within the same system. Table 1 uses different languages and observation units, precluding numerical

ranking. The purposive narrative review and unavailable full texts also limit coverage; overlapping erosion-study projects limit source independence.

Conclusion validity and next steps. Finite enumeration and small changed-item counts support bounded implementation checks. Population estimates need independent study units; speedup and scalability need paired workloads with controlled timing conditions. Broader evaluation should freeze projects and tools before predictions, label the full declared source scope, retain disagreements and report per-project errors, coverage and failures. The appendix describes the prepared D3 follow-up and the separate independent study requirements. Practitioner evaluation should measure false-block burden and review effort.

7 Conclusion

ArchSync integrates typed service-level observations, explicit conformance rules and evidence-backed Git decisions. The controlled evaluation demonstrates agreement across 20 development patches and incremental/full replay; the reliability campaign exercises the finite rule contract and repairs reproducible extraction and cache defects. Together, these artifacts provide a reproducible foundation for architecture checks at pull-request time. Independent repository evaluation and practitioner studies are the next steps. The integration gives maintainers an inspectable path from an introduced service interaction to a rule violation or an explicit architecture decision.

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A Reproduction and Interpretation Details

This appendix makes the development evidence inspectable through operational details for the single-column review profile and a separate supplement for the compact profile. Supplementary-upload acceptance requires confirmation from the venue.

A.1 Keep historical and current evidence separate

Historical D1/D2/P3 observations are reproduced from their pinned benchmark inputs and packaged analyzers. The reporting derivation uses benchmark commit `24d63ebf2fc3075a1d64f1eaff38cdc0b7f586fb`. Its ground truth, Phase 2 and Phase 3 records have individual hashes. Rerunning a current analyzer on that same corpus checks preservation of known development behavior; before/after reporting retains the original observations alongside these later regression results.

The artifact distinguishes original packages, current source snapshots, current distribution builds, first failing runs, repaired runs and unchanged historical result inputs. Current source and package hashes identify the exact evaluated candidate. The accompanying manifest maps each new manuscript count to its raw receipt; installation checks establish operational compatibility.

The historical evidence-location decomposition is explicitly post-hoc. Nine finding-bearing cases concern call sites; cases 08 and 17 concern deterministic anchors for missing edge/path requirements. Case 15 contains two violated rules but counts once. These matches measure the specified locations; completeness, ranking and developer comprehension require separate evaluation.

A.2 Finite oracle construction

For three nodes, six ordered non-self node pairs and two edge types give 12 binary edge slots, hence $2^{12} = 4096$ graphs. For each graph, an adjacency matrix is built independently for all edges and for each individual edge type. Direct requirements query that matrix; path requirements query its transitive closure. The diagonal starts false because a required path must contain an edge; a cycle can establish a nonempty path back to its starting node.

For each of four rule kinds, the source and target independently range over three exact identifiers and a full wildcard, and the edge-type restriction ranges over unrestricted and the two types. Thus $4 \times 4 \times 4 \times 3 = 192$ rule configurations are checked for every graph. The oracle compares per-rule violation counts, aggregate violations and classification.

Separately implemented graph-building, selector matching and traversal reduce direct implementation reuse. Shared rule interpretation and developer involvement remain sources of correlated error.

Ordering and consistent-renaming checks use masks 0 through 4080 in steps of 17, giving 241 graph states. Inverse-delta checks compare every graph with the empty-edge graph. They verify that added edges in one direction are removed edges in the other, and that edge changes without rules classify as evolution. Exhaustiveness applies to this declared domain; arbitrary graph pairs, metadata changes, wildcard fragments and source-language programs fall outside it.

A.3 Mutation procedure and retained failures

The mutation harness copies the relevant source and test into a temporary isolated workspace, changes one declared expression at a time, and invokes the same test suite with a fixed timeout. Production source remains unchanged by this procedure. Every run retains the replacement text, original and mutated source hashes, test hash, command output, exit state and native report. A passing unmodified baseline is required before mutant outcomes are interpreted.

Mutant states distinguish test passes, assertion failures and execution errors. The eight faults disable deny enforcement, invert allow logic, invert direct-require satisfaction, ignore path edge types, prevent traversal beyond one edge, miss path target detection, demote violation, or demote evolution. Compile errors and timeouts retain execution-error status. Sensitivity is assessed against these declared faults, with equivalent-mutant analysis outside the campaign's scope.

The original surviving classification mutation exposed a missing assertion on the returned decision despite correct finding counts. Adding that assertion detected the same fault. Both versions remain available, documenting the test-strengthening intervention and its effect on the declared fault model.

A.4 Detector repairs and residual uncertainty

The first adversarial set contains 18 cases: 15 failed before intervention and all 18 passed after it. Positive controls check aliases, captured resources and an unshadowed global fetch alongside local bindings. Type-only import tests deliberately include source that a TypeScript compilation gate would reject, exercising parser robustness when that gate is absent.

A later intervention added eight cases for concatenation and reassignment: seven failed in the cumulative 26-case pre-repair suite. Three further boundary controls were added after repair, yielding 29 passing adversarial cases. Reporting the changing suite sizes preserves the intervention sequence. An old positive fixture concatenated a suffix into an invalid URL port; the fixture was corrected to a valid path suffix and the invalid-port input retained as a negative regression.

A third intervention adds 14 cases to the 29-case suite: 12 write scenarios and two controls for unrelated or shadowed bindings. The identical 43-case suite has 11 failures and 32 passes before the change, then 43 passes afterward. One added write scenario already passed. The repair resolves variable identities within array/object destructuring (including rest/default targets), increment/decrement operations and assignment targets in for-in/for-of loops. For example, an array assignment that overwrites a URL binding must invalidate its initializer as the target of a later fetch. These constructed programs exercise specified behavior.

Recognized-write invalidation is conservative and file-wide, so it can suppress a genuine call before reassignment. Property mutation, imported constructor rebinding, dynamically built templates and alternative fallback destinations remain separate risks. Missing observations remain uncertain even when emitted evidence carries confidence constants. Analyzer version 0.4 invalidates caches from earlier semantics. The full repository gate passed from a clean temporary

candidate whose 410 tracked files matched the working source byte-for-byte. The first full run stopped at a CLI assertion still expecting version 0.3; that failure is retained alongside the corrected assertion and passing rerun as packaging and regression evidence.

A.5 Incremental and cache experiment boundary

The incremental suite creates temporary Git repositories and checks additions, modifications, deletions, renames within and between components, and complete-component deletion. Cold and warm results are compared with fresh full scans. The initial nine adversarial cases all failed on the original implementation; a later ignored-source case exposed a further false PASS. Before and after receipts preserve those observations.

Most differential cases compare classification, decisions, finding count and scanned-file count. Cold/warm comparisons additionally inspect architecture deltas and introduced/resolved findings. The nested-component case compares the complete observed graph and evidence. These comparison scopes distinguish aggregate agreement from edge identity and bound the tested Git configurations and filesystem layouts.

The baseline cache separates project-relative roots, architecture identity and analyzer semantics. A payload hash rejects accidentally changed JSON; concurrent writers use distinct temporary publication paths. Cache-writer authentication remains outside this integrity mechanism. Git inventory and scanning require a stable worktree throughout the check. Ignored source outside excluded build/dependency directories is included to match the full scan's source scope.

A.6 Exploratory D3 and independent follow-up

The prepared D3 packet is future exploratory work on 56 primary changes and four context-only changes from three public repositories, using direct cross-group source-file module dependencies. Both author reviewers are development-associated and previously exposed to tool output; their planned source reviews are nonblind, with any AI use recorded and source-verified. Reviews and paired tool runs remain pending; this manuscript reports no D3 results. Source-file endpoints preserve localization detail, while the module-only scope leaves service-level HTTP and database extraction for separate evaluation.

A separate independent follow-up requires a repository frame selected before prediction inspection, lineage-aware development/pilot/holdout separation, and reviewers independent of analyzer development and tuning. They must enumerate supported relations across the complete declared source scope, seal separate labels, and preserve disagreements before adjudication. Full-scope enumeration supplies the missed-relation denominator needed for recall; prediction-only checks address precision. Supported-vocabulary coverage, execution failures and unresolved labels require separate accounting.

A comparator must share a nonempty, frozen observation unit with ArchSync, preserving the distinction between import dependencies and remote service interactions. Performance experiments must pair the same workloads and machines, control the graph-cache condition separately from the operating-system disk cache, separate priming from timed execution and retain failures. These designs specify the proposed follow-up protocol.